

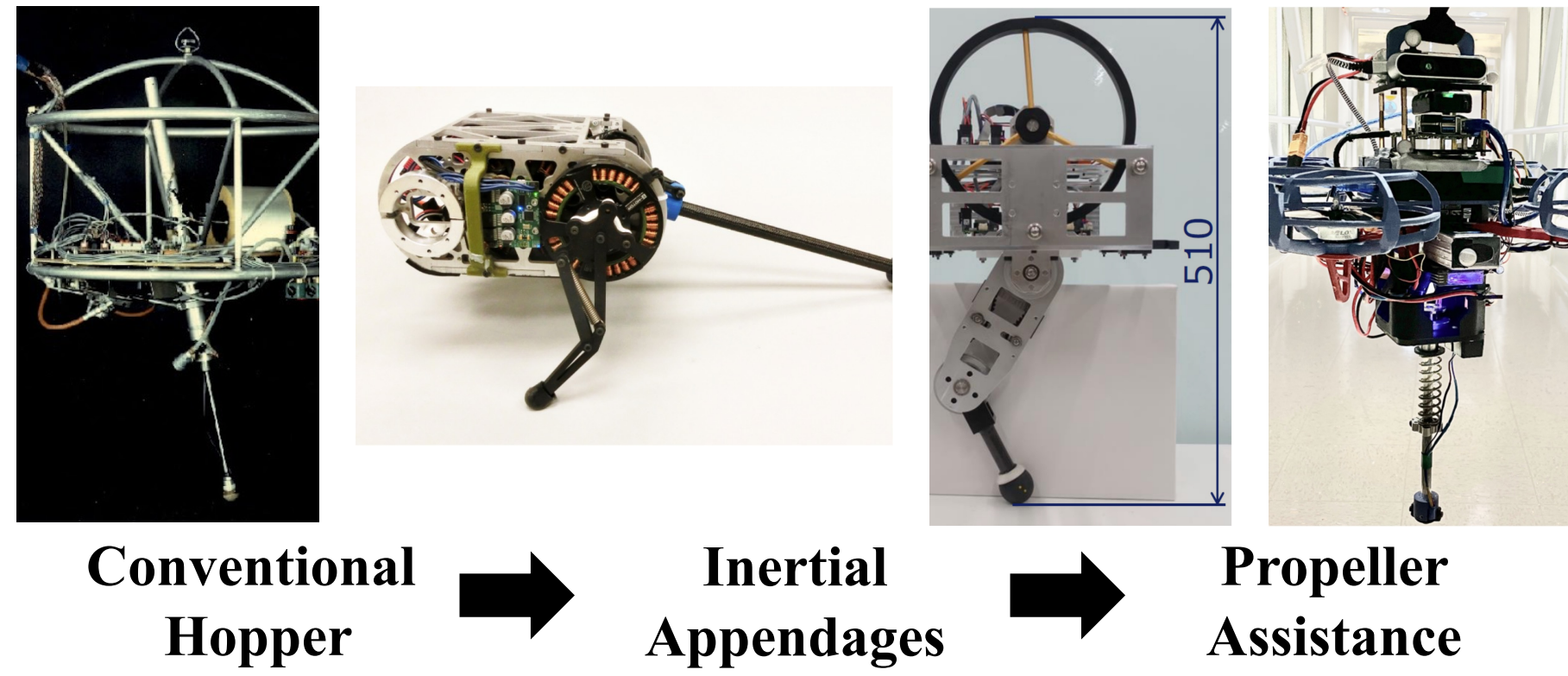
Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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1 Problem- hybrid dynamics create an authority gap

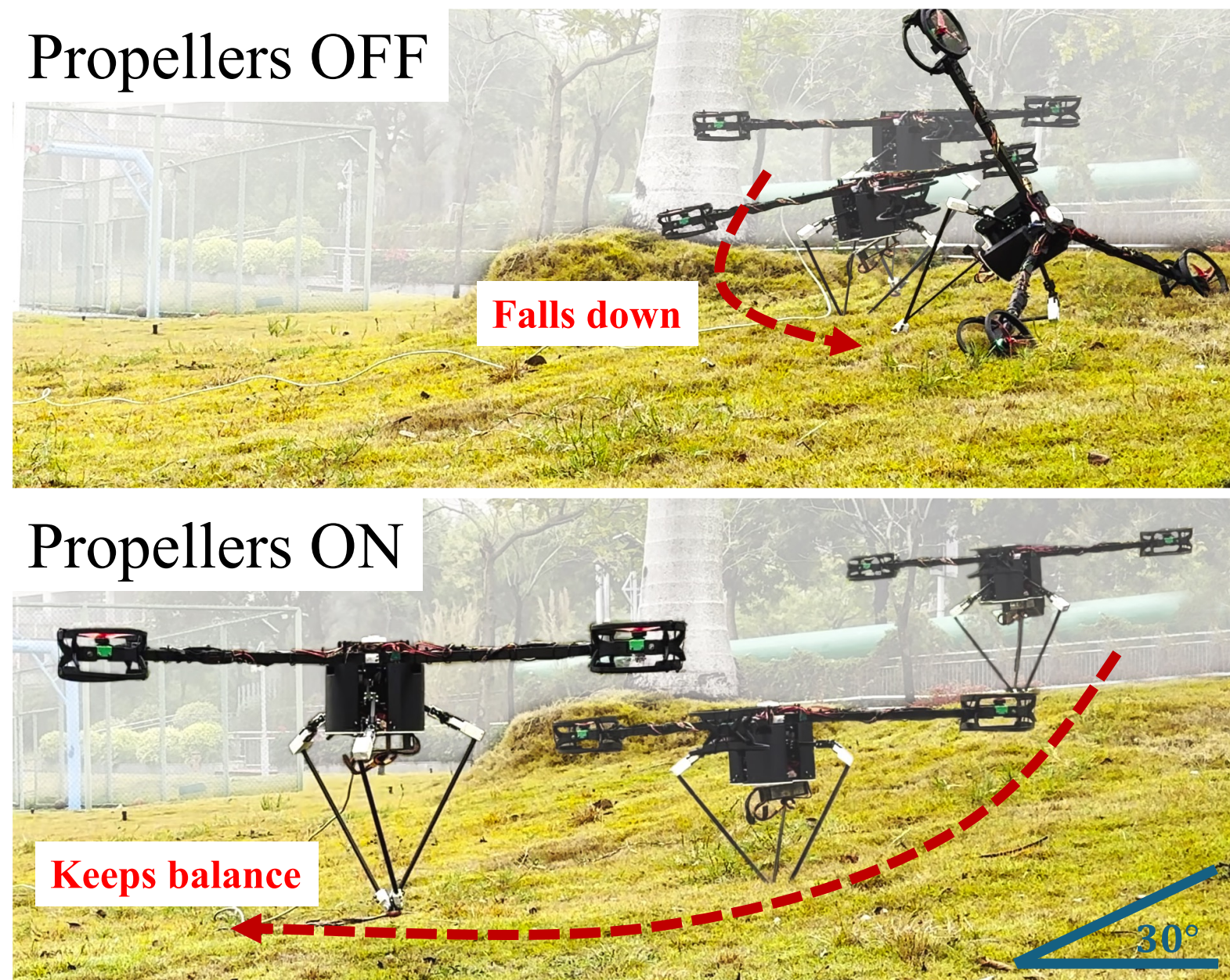


- **Stance**
Ground reaction force exists only during a short contact interval.
- **Flight**
No contact: attitude becomes underactuated.
- **Disturbance**
Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof- stable on 30° grass



- **Propellers off**
After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

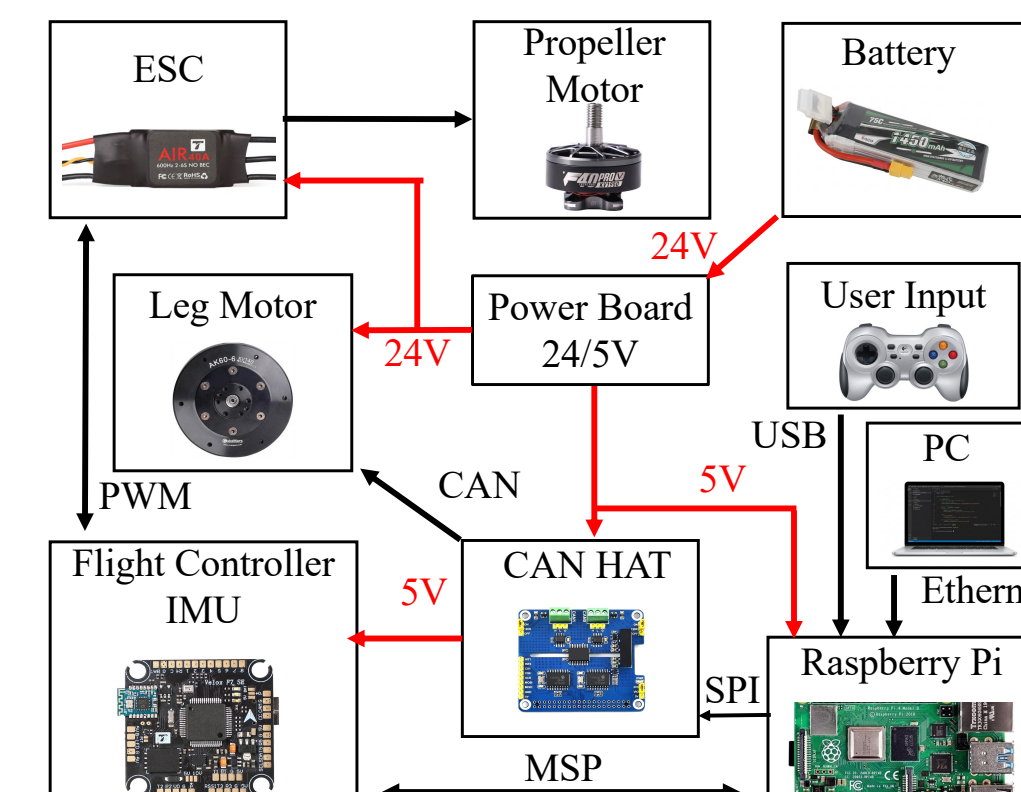
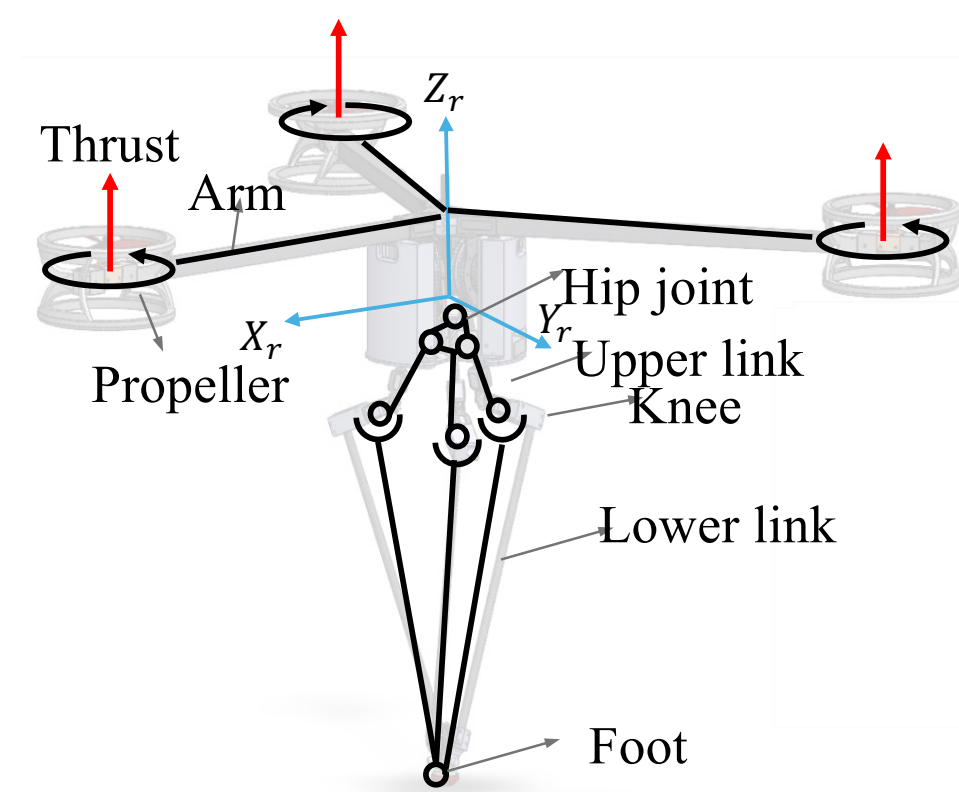
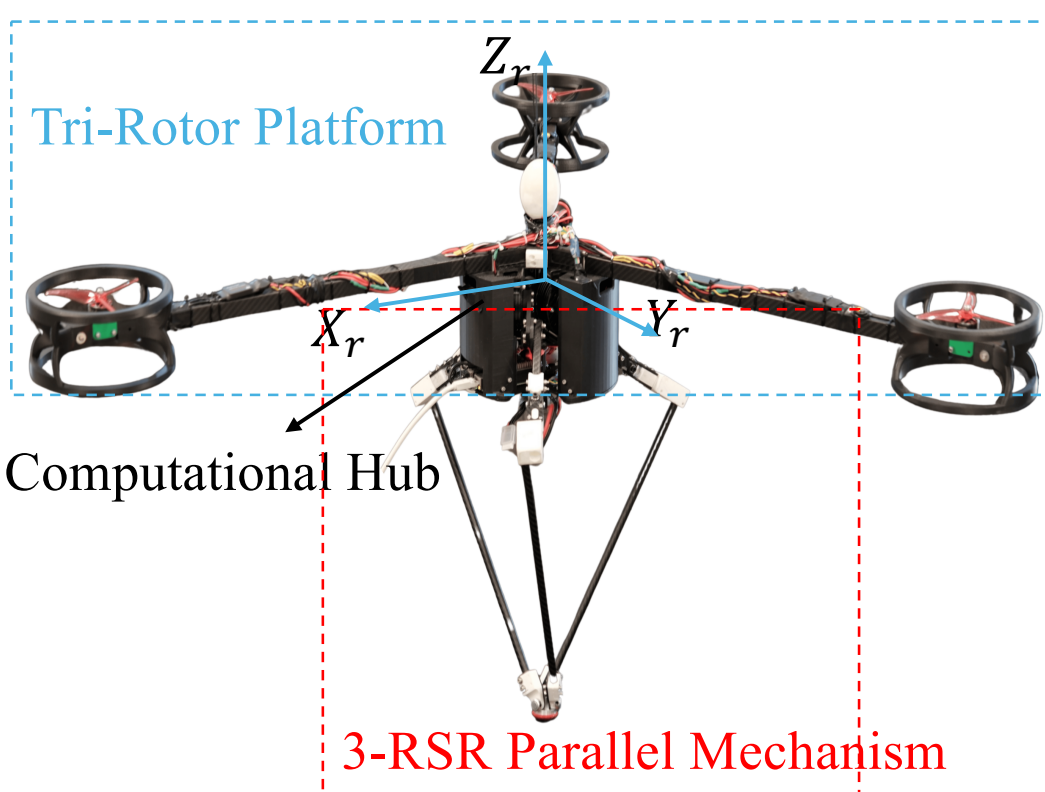
- **Propellers on**
Differential thrust remains available through flight and restores attitude before the next landing.

- **What this establishes**
Residual authority in stance.
Primary attitude authority in flight.
Faster recovery from uneven impacts.

Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3 Platform- hardware built for leg-first hopping



Robot design

Leg and tri-rotor components

Electronic architecture

- **3-RSR parallel leg**
Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

- **Computational hub**
Batteries and electronics near the hip: low center of mass, low inertia.

- **Y-layout tri-rotor**
Continuous attitude authority; **10 N** per rotor, total 0.82 mg: a hopper, not a drone.

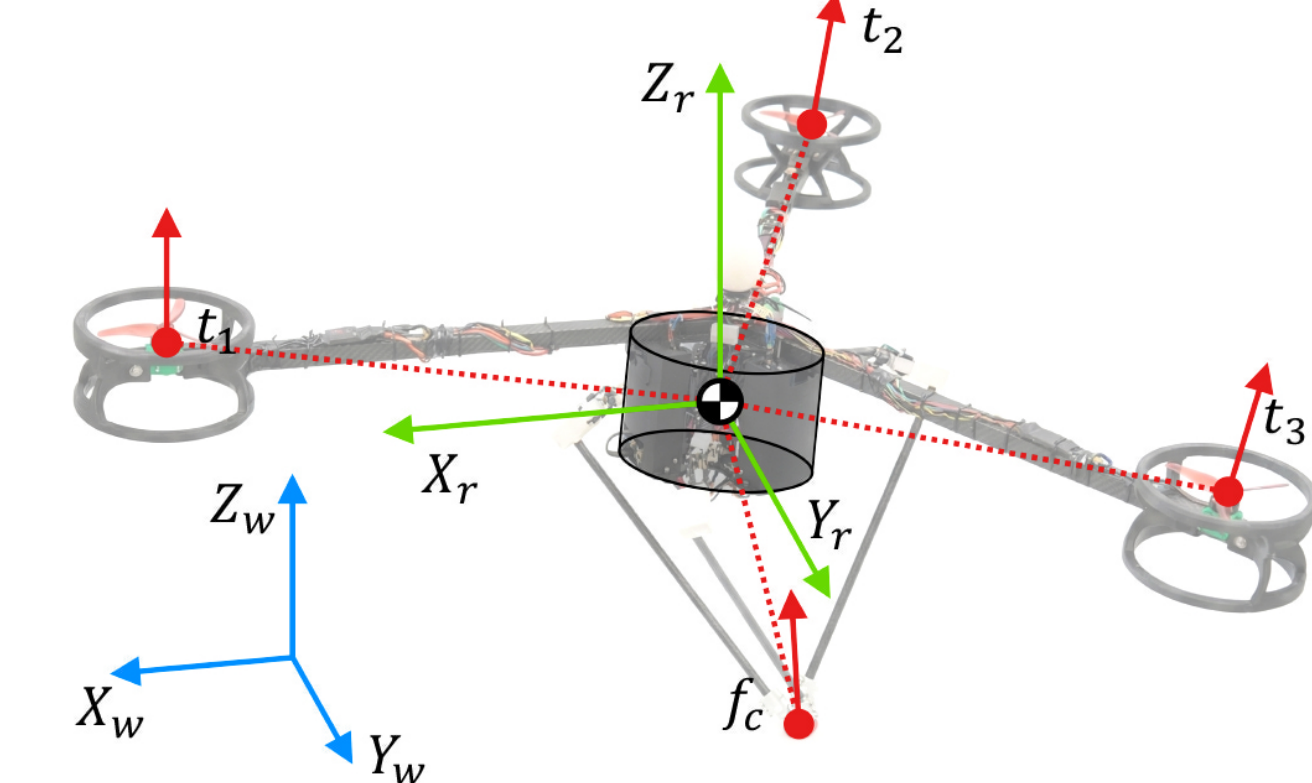
- **External PC @ 500 Hz**
State machine, estimation, wrench planning, and the HFA allocator.

- **Raspberry Pi interface**
CAN to the three leg motors; MSP to the flight controller.

- **Real-time link**
LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

3.75 kg total ■ 9.2% leg mass ■ 20 Nm /leg motor ■ 500 Hz control

4 Model- one rigid body, two actuator sets



Contact force f_c and thrusts $t_{1,2,3}$ produce the body wrench.

- **Net wrench = leg contact + three rotors**

$$\begin{bmatrix} F_{net} \\ \tau_{net} \end{bmatrix} = \begin{bmatrix} f_c \\ r_c \times f_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} d_i \\ r_i \times d_i \end{bmatrix} t_i$$

- **Newton-Euler dynamics of the CoM**

$$ma = F_{net} + mg, \quad \dot{\omega} = I^{-1}(R^T \tau_{net} - \omega \times I \omega)$$

- **Symbols (values used in all experiments)**

f_c : ground reaction force (world)

r_c, r_i : arms: CoM → contact, rotor i

$t_i \in [0, 10]$ N thrust along $d_i = Re_3$

F_{net}, τ_{net} : net force, moment

$m=3.75$ kg; g : gravity; I : inertia

$R \in SO(3)$ attitude; ω : body rate

J : leg Jacobian; $\tau_{max}=20$ Nm limit

$\mu=0.4$ friction coefficient

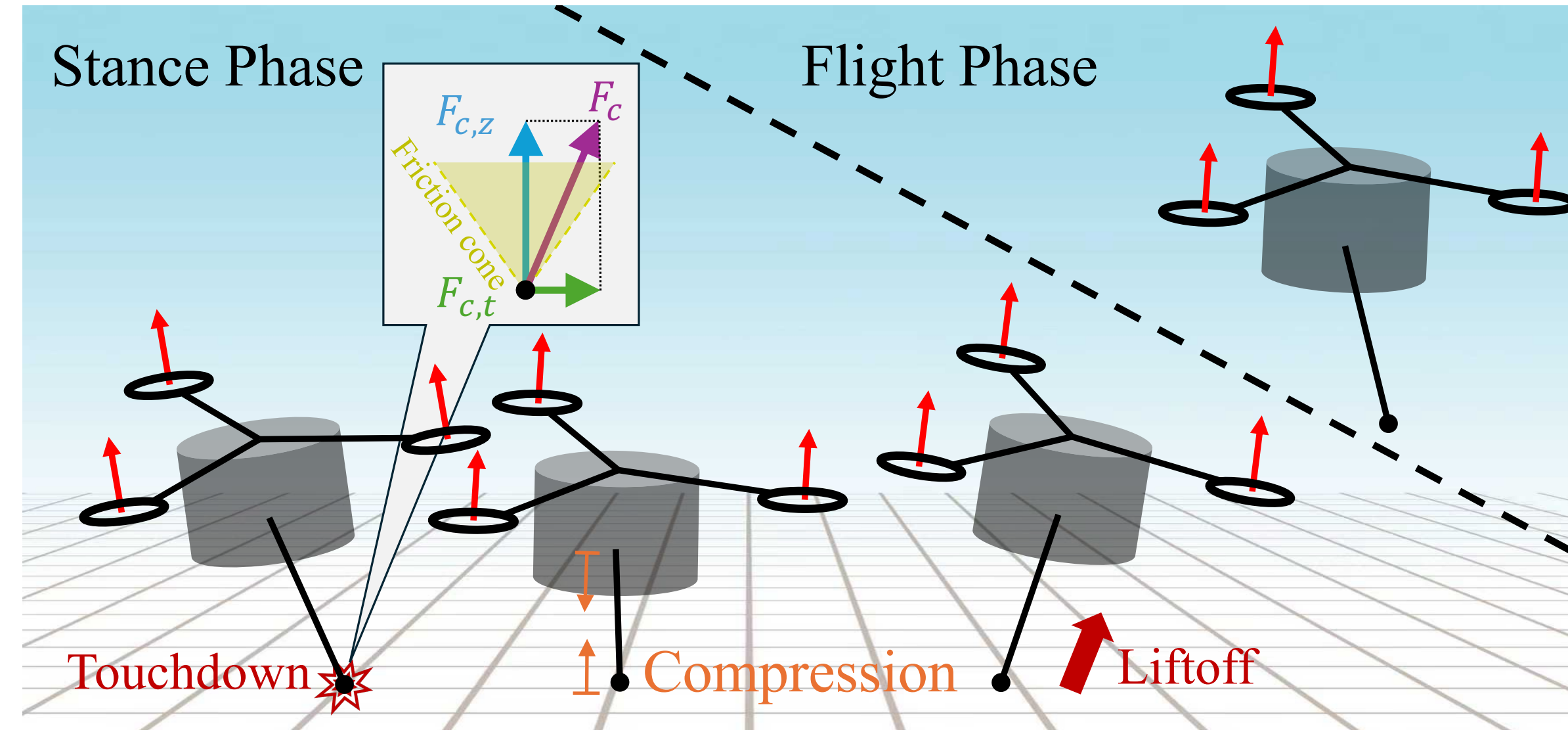
$p_z, \dot{p}_z$: CoM height, velocity; $h_{des}=0.664$ m

l : leg length; $l_0=0.464$ m; Δ_{td} : threshold

k_z, b_z, k_E : vertical, k_v, k_r : Raibert gains

K_R, K_ω : attitude PD gains

5 Onehop- event-driven authority handoff



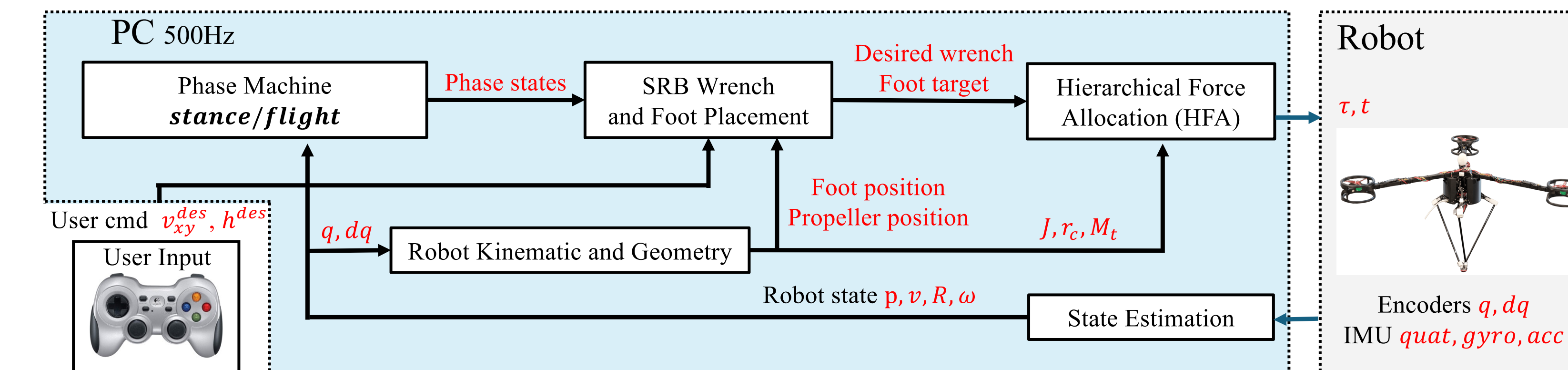
- **Touchdown:** $l \leq l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.

- **Liftoff:** $l \geq l_0$
Contact wrench vanishes; rotors primary for attitude.

- **Handoff**
Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

6 Control- leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

- **A. Vertical energy hopping (stance)**

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + mg p_z, \quad E_{des} = mg h_{des}$$

$$f_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

- **B. SO(3) attitude PD (both phases)**

$$e_R = \frac{1}{2} (R_{des}^T R - R^T R_{des})^V$$

$$\tau_b^{des} = -K_R e_R + K_\omega (\omega^{des} - \omega)$$

Roll/pitch only; yaw left free. $\tau^{des} = R \tau_b^{des}$.

- **C. Leg first- constrained least squares**

$$f_{c,xy}^{cmd} = \arg \min_{f_{xy}} \|(r_c \times f_c)_{rp} - (\tau^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|f_{xy}\|_2 \leq \mu f_{c,z}^{des}, \quad \|J^T(-R^T f_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

- **D. Rotors take only the residual**

$$\tau_{res}^{cmd} = \tau^{des} - r_c \times f_c^{cmd}$$

$$\delta t^{cmd} = \arg \min_{\delta t} \|(M_t \delta t)_{rp} - (\tau_{res}^{cmd})_{rp}\|_2^2$$

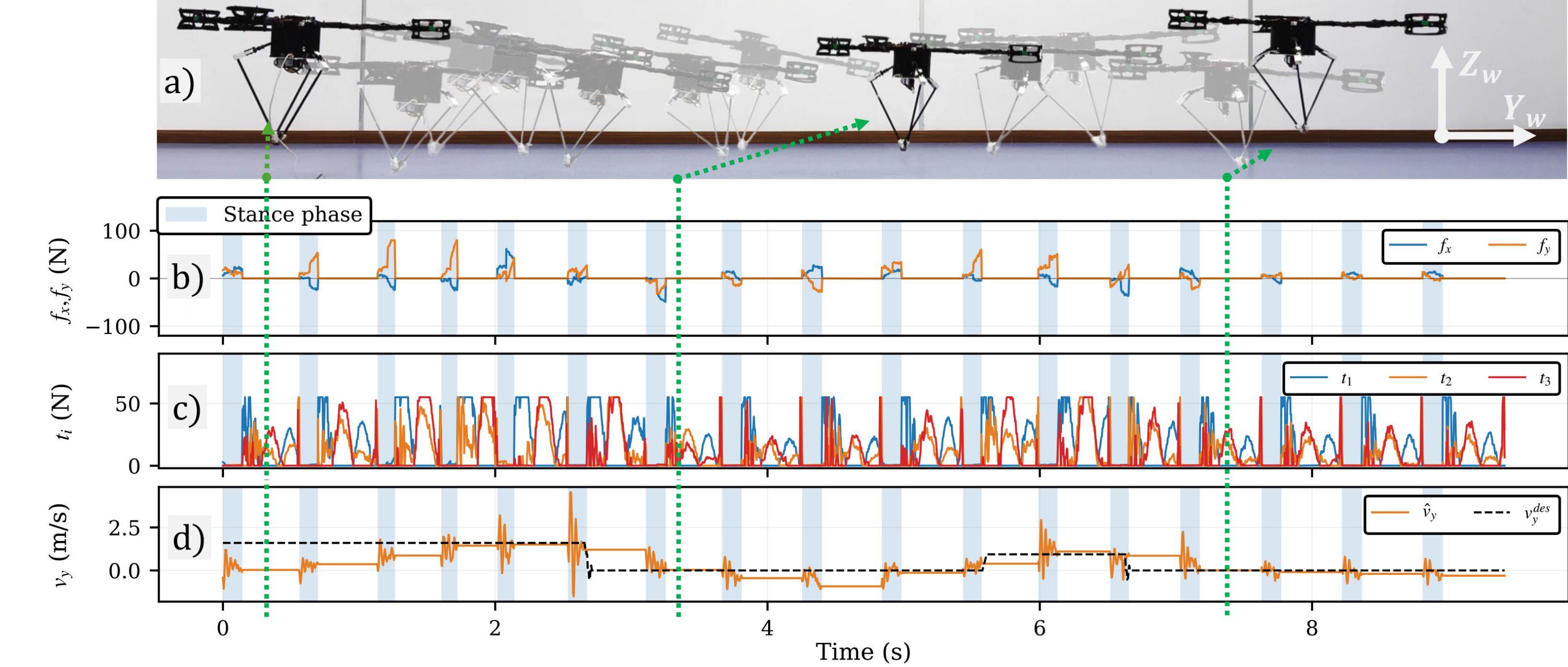
s.t. thrust bounds; $t^{cmd} = \frac{T_{base}}{3} 1_3 + \delta t^{cmd}, T_{base} = 0.02$ mg.

- **E. Flight- rotors primary; Raibert-style foot placement**

$$f_c = 0; \quad \text{rotors track } (\tau^{des})_{rp}; \quad \tau_{td,xy}^{des} = k_v v_{xy} - k_r v_{xy}$$

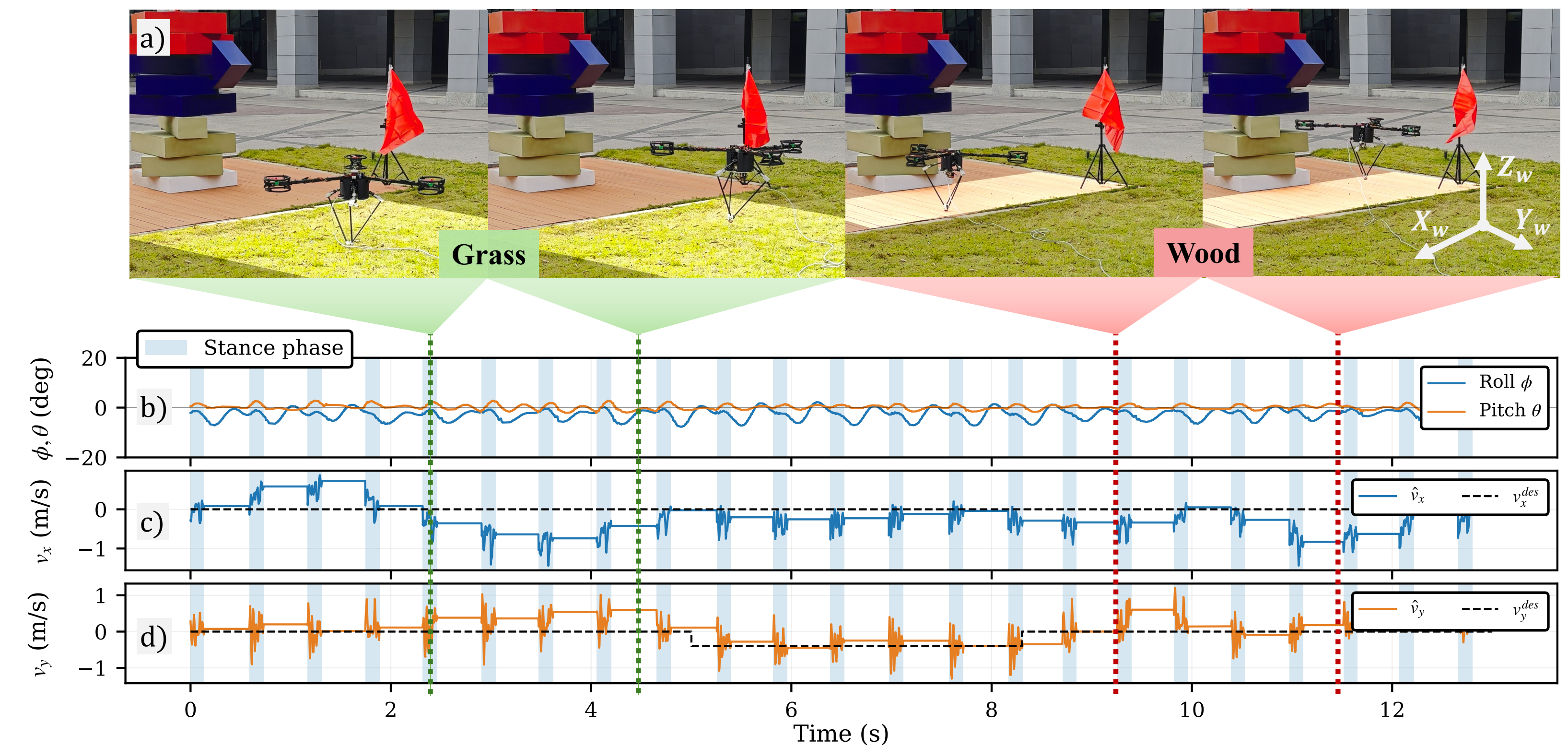
$$p_{foot}^{des} = [x_{td}^{des}, y_{td}^{des}, -\sqrt{l_0^2 - \|r_{td,xy}^{des}\|_2^2}]^T$$

7 Validation- indoor stop-and-go tracking



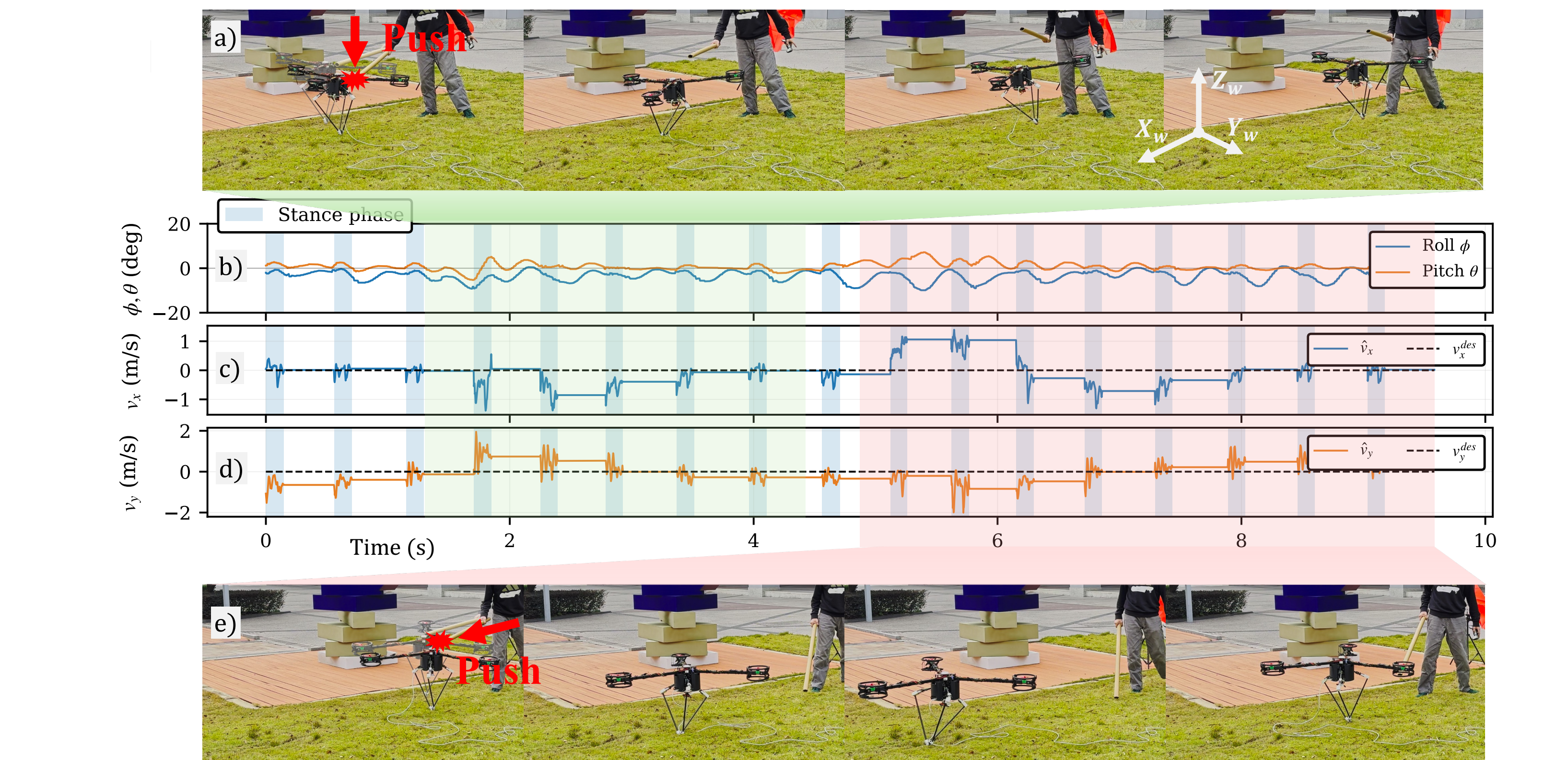
Stop-and-go over **17 hops**: velocity recovered stride to stride by **Raibert foot placement**.

8 Validation- grass to wood



Grass → wood, **23 hops**: same controller - no terrain model, no gain retuning.

9 Validation- push recovery



Rotor-arm and body pushes rejected - stable tracking again in **1-3 hops**.